ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

TECHNICAL SCIENCES P1

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Technical Sciences Paper 1 practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing or wrong units. Give follow-through marks for correct use of an earlier wrong answer.

QUESTION 1

1 1.1 B ✓✓ 1.2 B ✓✓ 1.3 C ✓✓ 1.4 B ✓✓ 1.5 C ✓✓ [10]

QUESTION 2

2.1 (a) kilogram (kg) ✓ (b) second (s) ✓ (c) ampere (A) ✓ (3)

2.2 $(a)\ 2\,500\ \text{N}$ ✓ $(b)\ 0{,}45\ \text{m}$ ✓ $(c)\ 72 \div 3{,}6$ ✓ $= 20\ \text{m·s}^{-1}$ ✓ (4)

2.3 $3{,}2$ ✓ $\times\ 10^{-4}\ \text{m}$ ✓ (2)

2.4 $28 \times 0{,}01 = 0{,}28\ \text{mm}$ ✓; $5{,}5 + 0{,}28$ ✓ $= 5{,}78\ \text{mm}$ ✓ (3)

[12]

QUESTION 3

3.1.1 $12 + 5 = 17\ \text{km}$ ✓ (1)

3.1.2 $12 - 5 = 7\ \text{km}$ ✓ north ✓ (2)

3.1.3 $20\ \text{min} = \frac{1}{3}\ \text{h}$ ✓; $17 \div \frac{1}{3}$ ✓ $= 51\ \text{km·h}^{-1}$ ✓ (3)

3.1.4 $7 \div \frac{1}{3}$ ✓ $= 21\ \text{km·h}^{-1}$ ✓ north ✓ (3)

3.2.1 The rate of change of velocity ✓✓. (2)

3.2.2 $30 = 10 + a(5)$ ✓✓ $\rightarrow a = 4\ \text{m·s}^{-2}$ ✓ (3)

3.2.3 $\Delta x = \frac{10 + 30}{2} \times 5$ ✓✓ $= 100\ \text{m}$ ✓ (3)

3.3 A scalar has magnitude only, e.g. distance, speed, time ✓; a vector has magnitude and direction ✓, e.g. displacement, velocity, force ✓. (3)

[20]

QUESTION 4

4.1 $w = mg = 200 \times 9{,}8$ ✓ $= 1\,960\ \text{N}$ ✓ (2)

4.2 $W = F\Delta x = 1\,960 \times 3$ ✓✓ $= 5\,880\ \text{J}$ ✓ (3)

4.3 Power is the rate at which work is done (energy is transferred) ✓✓. $P = \frac{5\,880}{12}$ ✓✓ $= 490\ \text{W}$ ✓ (5)

4.4 $E_P = mgh = 5\,880\ J$ ✓✓ (2)

4.5 $P_{in} = \frac{490}{0,7}$ ✓✓ $= 700\ W$ ✓ (3)

4.6 Energy cannot be created or destroyed; it can only be changed from one form to another ✓✓. Motor: electrical energy → mechanical (kinetic) energy ✓. (3)

[18]

QUESTION 5

5.1 Longitudinal: particles vibrate parallel to the direction of the wave ✓, e.g. sound ✓. Transverse: particles vibrate perpendicular to the direction of the wave ✓, e.g. waves on a rope, water waves, light ✓. (4)

5.2 $v = f\lambda = 200 \times 1,7$ ✓✓ $= 340\ m{\cdot}s^{-1}$ ✓ (3)

5.3 Ultrasound pulses are sent into the metal; a crack reflects the waves back early ✓, and the time of the echo shows where the crack is ✓. (2)

5.4 Very loud sound (about 85 dB and above) ✓ over long periods damages the hair cells in the inner ear ✓, causing permanent hearing loss ✓. (3)

[12]

QUESTION 6

6.1 The opposition to the flow of electric current in a conductor ✓✓ (ratio of potential difference to current). (2)

6.2 $\frac{1}{R_p} = \frac{1}{6} + \frac{1}{12}$ ✓ $= \frac{3}{12}$ ✓ $\rightarrow R_p = 4\ \Omega$ ✓ (3)

6.3 $2 + 4 = 6\ \Omega$ ✓ (1)

6.4 $I = \frac{V}{R} = \frac{12}{6}$ ✓✓ $= 2\ A$ ✓ (3)

6.5 $V = IR = 2 \times 2$ ✓ $= 4\ V$ ✓ (2)

6.6 $V_{parallel} = 12 - 4 = 8\ V$ ✓; $I = \frac{8}{12}$ ✓ $= 0,67\ A$ ✓ (3)

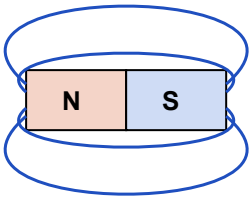
6.7 $P = I^2R = 2^2 \times 2$ ✓✓ $= 8\ W$ ✓ (3)

6.8 A fuse is a thin wire connected in series with the appliance ✓. If the current becomes too high, the wire melts ✓ and breaks the circuit, preventing damage, overheating or fire ✓. (3)

[20]

QUESTION 7

7.1 (3)



Marks: closed field lines from N around to S ✓; lines do not cross and are closest together at the poles ✓; arrows from N to S outside the magnet ✓

7.2 Any TWO: increase the current ✓; increase the number of turns of the coil ✓; use a soft-iron core (2)

7.3 Any ONE: lifting scrap metal (crane) ✓ / relays and switches / electric bells / motors / separating iron from other waste (1)

7.4 Any TWO: iron ✓; nickel ✓; cobalt; steel (2)

[8]

TOTAL: 100

END OF MARKING GUIDELINE